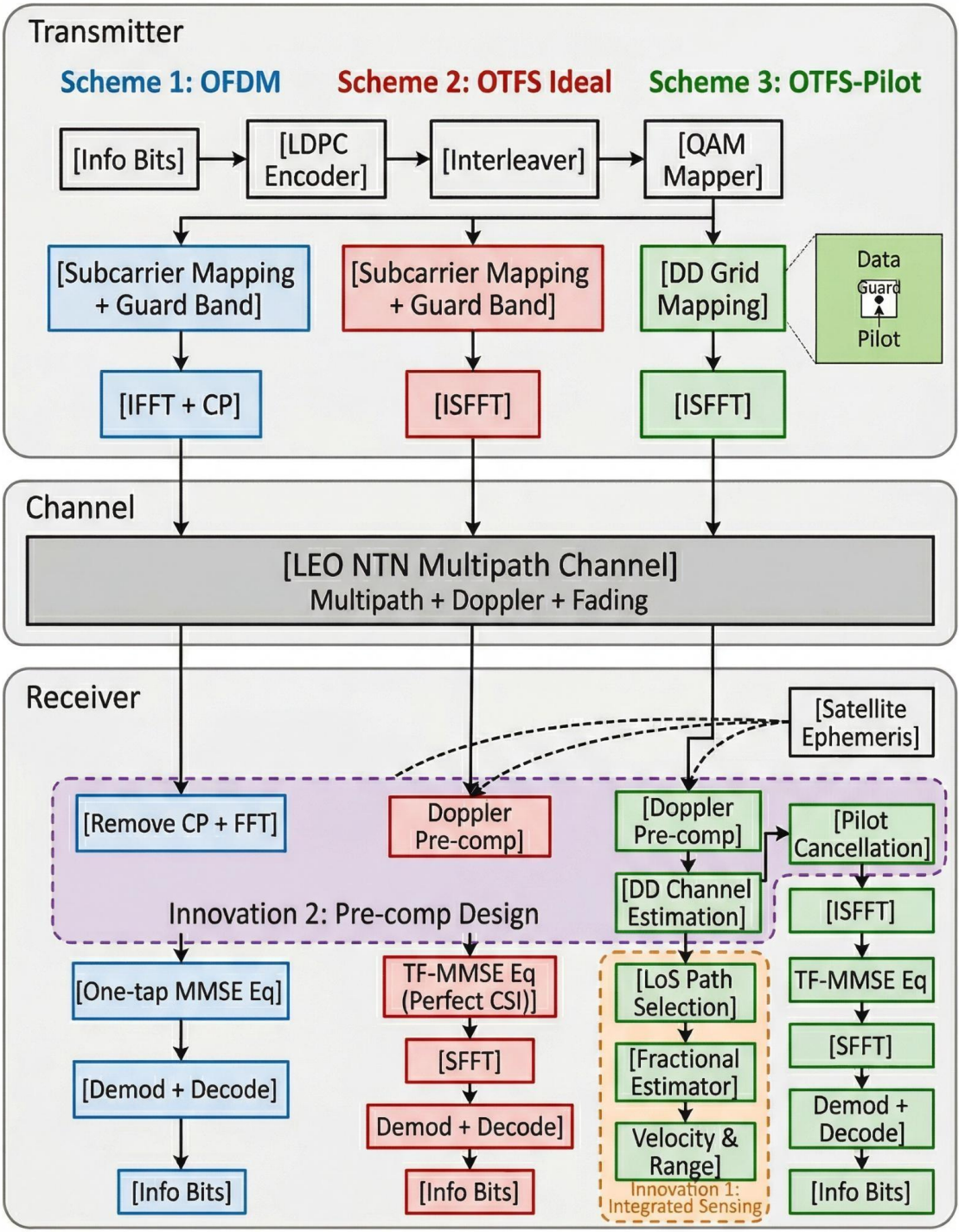
Transmitter
Scheme 1: OFDM
Scheme 2: OTFS Ideal
Scheme 3: OTFS-Pilot
[Info Bits]
[LDPC Encoder]
[Interleaver]
[QAM Mapper]
[Subcarrier Mapping + Guard Band]
[Subcarrier Mapping + Guard Band]
[DD Grid Mapping]
Data
Guard
Pilot
[IFFT + CP]
[ISFFT]
[ISFFT]
Channel
[LEO NTN Multipath Channel]
Multipath + Doppler + Fading
Receiver
[Satellite Ephemeris]
[Remove CP + FFT]
Doppler Pre-comp]
[Doppler Pre-comp]
[Pilot Cancellation]
[DD Channel Estimation]
Innovation 2: Pre-comp Design
[ISFFT]
[One-tap MMSE Eq]
TF-MMSE Eq (Perfect CSI)]
[LoS Path Selection]
TF-MMSE Eq
[SFFT]
[Fractional Estimator]
[SFFT]
[Demod + Decode]
Demod + Decode]
Velocity & Range]
Demod + Decode]
Innovation 1: Integrated Sensing
[Info Bits]
[Info Bits]
[Info Bits]